

Lecture 18

Production Functions

ECO303

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Marginal Productivity

- The firm's production function
 - For a particular good (q)
 - Shows the maximum amount of the good that can be produced
 - Using alternative combinations of capital (k) and labor (l)

$$q = f(k, l)$$

Marginal Physical Product

- Marginal physical product
 - The additional output that can be produced
 - By employing one more unit of that input
 - Holding other inputs constant

$$\text{marginal physical product of capital} = MP_k = \frac{\partial q}{\partial k} = f_k$$

$$\text{marginal physical product of labor} = MP_l = \frac{\partial q}{\partial l} = f_l$$

Diminishing Marginal Productivity

- Marginal physical product
 - Depends on how much of that input is used
- Diminishing marginal productivity

$$\frac{\partial MP_k}{\partial k} = \frac{\partial^2 f}{\partial k^2} = f_{kk} = f_{11} < 0$$

$$\frac{\partial MP_l}{\partial l} = \frac{\partial^2 f}{\partial l^2} = f_{ll} = f_{22} < 0$$

Diminishing Marginal Productivity

- Changes in the marginal productivity of labor
 - Also depend on changes in other inputs such as capital
 - We need to consider f_{lk} which is often > 0

$$f_{lk} = \frac{\partial MP_l}{\partial k}$$

Average Physical Product

- Labor productivity
 - Often means average productivity
- Average product of labor

$$AP_l = \frac{\text{output}}{\text{labor input}} = \frac{q}{l} = \frac{f(k, l)}{l}$$

- AP_l also depends on the amount of capital employed

EXAMPLE 9.1 A Two-Input Production Function

- Suppose the production function for widgets can be represented by

$$q = f(k, l) = 600k^2l^2 - k^3l^3$$

- To construct MP_l and AP_l , we must assume a value for k
 - Let $k = 10$
- The production function becomes

$$q = 60,000l^2 - 1000l^3$$

EXAMPLE 9.1 A Two-Input Production Function

- The marginal productivity function is

$$MP_l = \partial q / \partial l = 120,000l - 3000l^2$$

- Which diminishes as l increases
- This implies that q has a maximum value:

$$120,000l - 3000l^2 = 0$$

$$40l = l^2$$

$$l = 40$$

- Labor input beyond $l = 40$ reduces output

EXAMPLE 9.1 A Two-Input Production Function

- To find average productivity, we hold $k=10$ and solve

$$AP_l = q/l = 60,000l - 1000l^2$$

- AP_l reaches its maximum where

$$\partial AP_l / \partial l = 60,000 - 2000l = 0$$

$$l = 30$$

- When $l = 30$, $AP_l = MP_l = 900,000$
 - When AP_l is at its maximum, AP_l and MP_l are equal

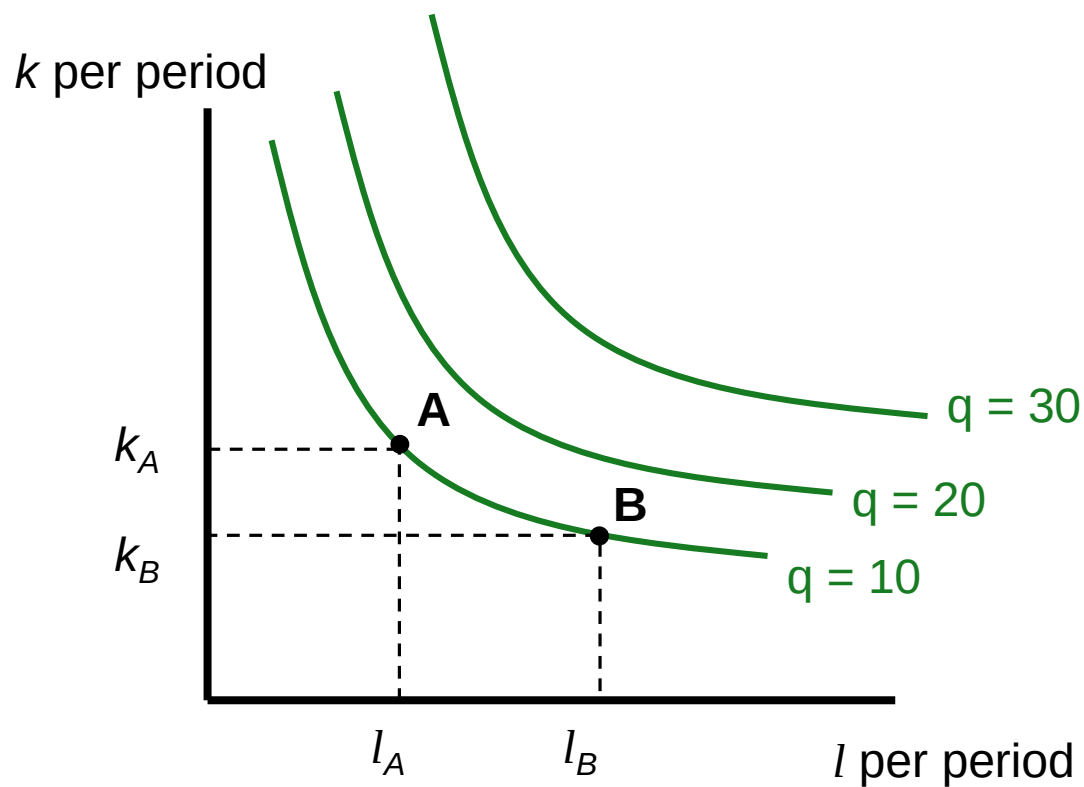
Isoquant Maps

- Isoquant map
 - To illustrate the possible substitution of one input for another
- An isoquant
 - Shows those combinations of k and l that can produce a given level of output (q_0)

$$f(k, l) = q_0$$

FIGURE 9.1

An Isoquant Map



Isoquants record the alternative combinations of inputs that can be used to produce a given level of output. The slope of these curves shows the rate at which l can be substituted for k while keeping output constant. The negative of this slope is called the (marginal) rate of technical substitution (RTS). In the figure, the RTS is positive and diminishing for increasing inputs of labor.

Marginal Rate of Technical Substitution

- Marginal rate of technical substitution (*RTS*)
 - Shows the rate at which labor can be substituted for capital
 - Holding output constant along an isoquant

$$RTS \text{ (} l \text{ for } k) = \left. \frac{-dk}{dl} \right|_{q=q_0}$$

RTS and Marginal Productivities

- Total differential of the production function:

$$dq = \frac{\partial f}{\partial l} \cdot dl + \frac{\partial f}{\partial k} \cdot dk = MP_l \cdot dl + MP_k \cdot dk$$

- Along an isoquant $dq = 0$, so

$$MP_l \cdot dl = -MP_k \cdot dk$$

$$RTS \text{ (} l \text{ for } k \text{)} = \left. \frac{-dk}{dl} \right|_{q=q_0} = \frac{MP_l}{MP_k}$$

RTS and Marginal Productivities

- *RTS* will be positive (or zero)
 - Because MP_l and MP_k will both be nonnegative
- Not possible to derive a diminishing *RTS*
 - From the assumption of diminishing marginal productivity alone
- To show that isoquants are convex
 - Show that $d(RTS)/dl < 0$

RTS and Marginal Productivities

- Since $RTS = f_l / f_k$
 - And $dk/dl = -f_l / f_k$ along an isoquant and Young's theorem ($f_{kl} = f_{lk}$)

$$\begin{aligned}\frac{dRTS}{dl} &= \frac{d(f_l / f_k)}{dl} \\ &= \frac{[f_k(f_{ll} + f_{lk} \times dk / dl) - f_l(f_{kl} + f_{kk} \times dk / dl)]}{(f_k)^2} \\ &= \frac{(f_k^2 f_{ll} - 2 f_k f_l f_{kl} + f_l^2 f_{kk})}{(f_k)^3}\end{aligned}$$

RTS and Marginal Productivities

- Denominator is positive
 - Because we have assumed $f_k > 0$
- The ratio will be negative if f_{kl} is positive
 - Because f_{ll} and f_{kk} are both assumed to be negative
- Intuitively, it seems reasonable that $f_{kl} = f_{lk}$ should be positive
 - If workers have more capital, they will be more productive

RTS and Marginal Productivities

- But some production functions have $f_{kl} < 0$ over some input ranges
 - Assuming diminishing *RTS* means that MP_l and MP_k diminish quickly enough to compensate for any possible negative cross-productivity effects

EXAMPLE 9.2 A Diminishing RTS

- Production function:

$$q = f(k, l) = 600k^2l^2 - k^3l^3$$

- Marginal productivity functions:

$$MP_l = f_l = 1200k^2l - 3k^3l^2$$

$$MP_k = f_k = 1200kl^2 - 3k^2l^3$$

- Will be positive for values of k and l for which $kl < 400$

EXAMPLE 9.2 A Diminishing RTS

- Because

$$f_{ll} = 1200k^2 - 6k^3l \text{ and } f_{kk} = 1200l^2 - 6kl^3$$

- Diminishing marginal productivities for sufficiently large values of k and l

$$f_{ll} \text{ and } f_{kk} < 0 \text{ if } kl > 200$$

- Cross differentiation of either of the marginal productivity functions yields

$$f_{kl} = f_{lk} = 2400kl - 9k^2l^2$$

- Which is positive only for $kl < 266$

Returns to Scale

- How does output respond to increases in all inputs together?
 - Suppose that all inputs are doubled, would output double?
- As inputs are doubled
 - Greater division of labor and specialization of function
 - Loss in efficiency - management may become more difficult

Returns to Scale

- Production function is given by $q = f(k, l)$
 - And all inputs are multiplied by the same positive constant ($t > 1$)
 - Then we classify the returns to scale of the production function by

Effect on Output	Returns to Scale
$f(tk, tl) = tf(k, l) = tq$	Constant
$f(tk, tl) < tf(k, l) = tq$	Decreasing
$f(tk, tl) > tf(k, l) = tq$	Increasing

Returns to Scale

- Production function
 - Constant returns to scale for some levels of input usage
 - Increasing or decreasing returns for other levels
 - The degree of returns to scale is generally defined within a fairly narrow range of variation in input usage

Constant Returns to Scale

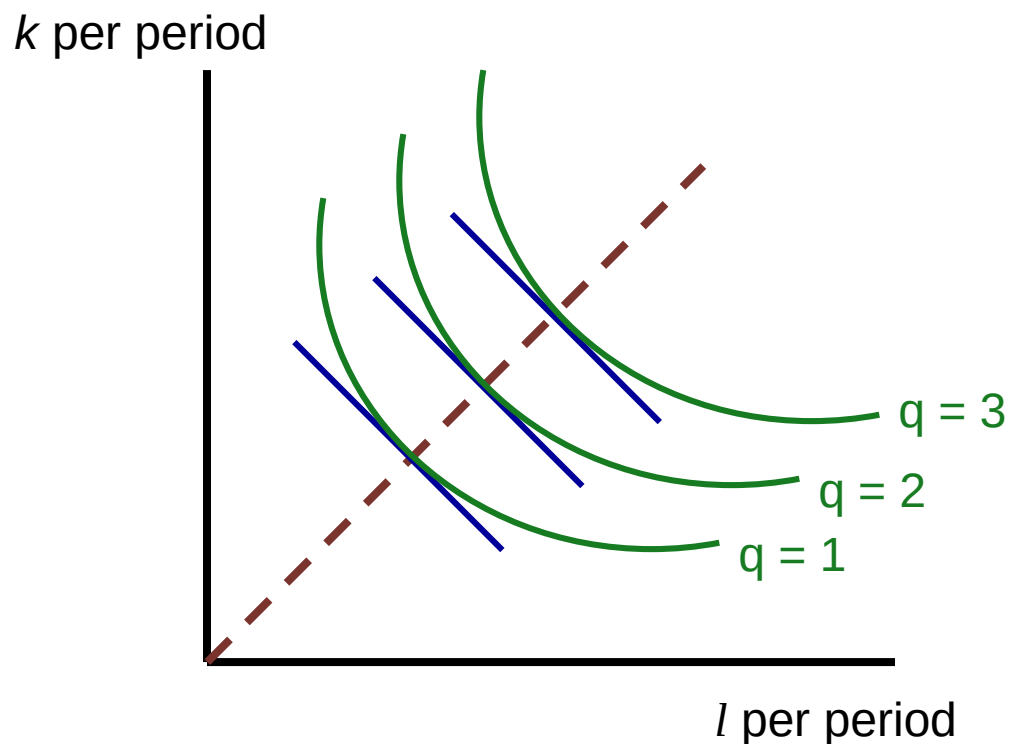
- Constant returns-to-scale production functions
 - Are homogeneous of degree one in inputs
$$f(tk,tl) = t^1 f(k,l) = tq$$
- The marginal productivity functions
 - Are homogeneous of degree zero
 - If a function is homogeneous of degree k , its derivatives are homogeneous of degree $k-1$

Constant Returns to Scale

- Marginal productivity of any input
 - Depends on the ratio of capital and labor
 - Not on the absolute levels of these inputs
- The *RTS* between k and l
 - Depends only on the ratio of k to l
 - Not the scale of operation
 - Homothetic production function
 - All of the isoquants are radial expansions of one another

FIGURE 9.2

Isoquant Map for a Constant Returns-to-Scale Production Function



Because a constant returns-to-scale production function is homothetic, the RTS depends only on the ratio of k to l , not on the scale of production. Consequently, along any ray through the origin (a ray of constant k/l), the RTS will be the same on all isoquants. An additional feature is that the isoquant labels increase proportionately with the inputs.

Returns to Scale

- Returns to scale can be generalized to a production function with n inputs

$$q = f(x_1, x_2, \dots, x_n)$$

- If all inputs are multiplied by a positive constant t :

$$f(tx_1, tx_2, \dots, tx_n) = t^k f(x_1, x_2, \dots, x_n) = t^k q$$

- If $k = 1$, constant returns to scale
- If $k < 1$, decreasing returns to scale
- If $k > 1$, increasing returns to scale

The Elasticity of Substitution

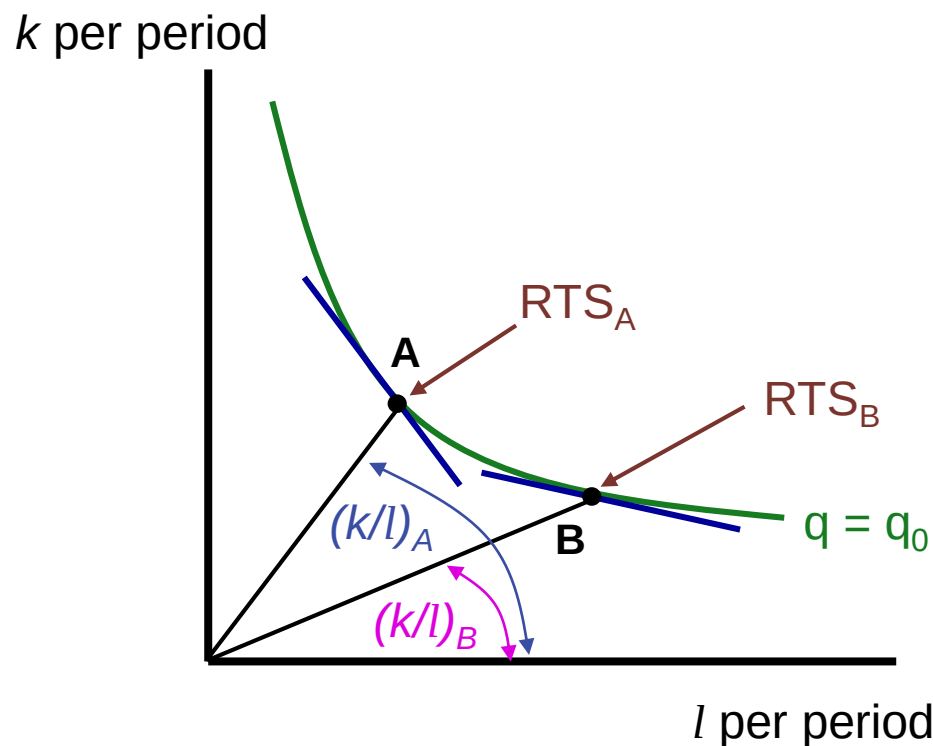
- Elasticity of substitution (σ)
 - For the production function $q = f(k, l)$
 - Measures the proportionate change in k/l relative to the proportionate change in the RTS along an isoquant

$$\sigma = \frac{\% \Delta(k/l)}{\% \Delta RTS} = \frac{d(k/l)}{dRTS} \times \frac{RTS}{k/l} = \frac{d \ln(k/l)}{d \ln RTS} = \frac{d \ln(k/l)}{d \ln(f_l / f_k)}$$

- The value of σ will always be positive because k/l and RTS move in the same direction

FIGURE 9.3

Graphic Description of the Elasticity of Substitution



In moving from point A to point B on the $q = q_0$ isoquant, both the capital–labor ratio (k/l) and the RTS will change. The elasticity of substitution (σ) is defined to be the ratio of these proportional changes; it is a measure of how curved the isoquant is.

Elasticity of Substitution

- If σ is high
 - The *RTS* will not change much relative to k/l
 - The isoquant will be relatively flat
- If σ is low
 - The *RTS* will change by a substantial amount as k/l changes
 - The isoquant will be sharply curved
- σ can change along an isoquant
 - Or as the scale of production changes

Elasticity of Substitution

- Elasticity of substitution between two inputs
 - The proportionate change in the ratio of the two inputs
 - To the proportionate change in RTS
 - With output and the levels of other inputs constant

The Linear Production Function

- Linear production function ($\sigma = \infty$):

$$q = f(k, l) = \alpha k + \beta l$$

- Constant returns to scale

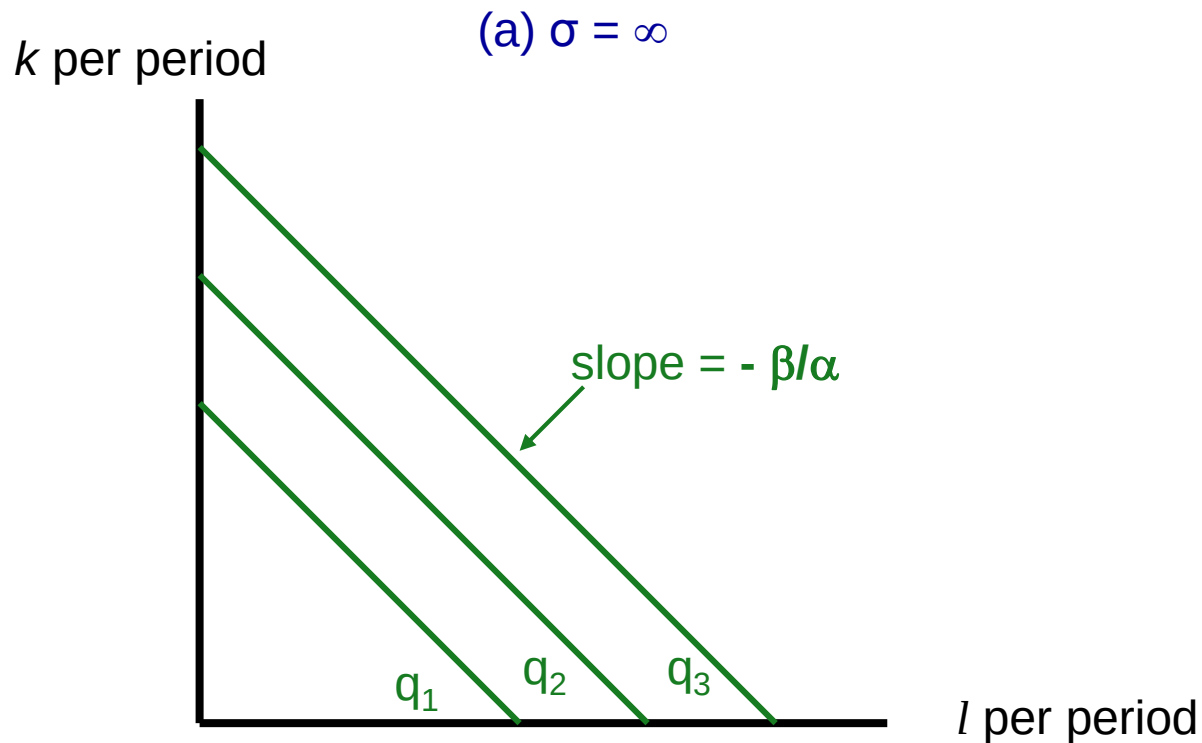
$$f(tk, tl) = \alpha tk + \beta tl = t(\alpha k + \beta l) = tf(k, l)$$

- All isoquants are straight lines with slope -
- α/β

- RTS is constant
- $\sigma = \infty$

FIGURE 9.4 (a)

Isoquant Maps for Simple Production Functions with Various Values for σ



Three possible values for the elasticity of substitution are illustrated in these figures. In (a), capital and labor are perfect substitutes. In this case, the RTS will not change as the capital–labor ratio changes.

Fixed Proportions

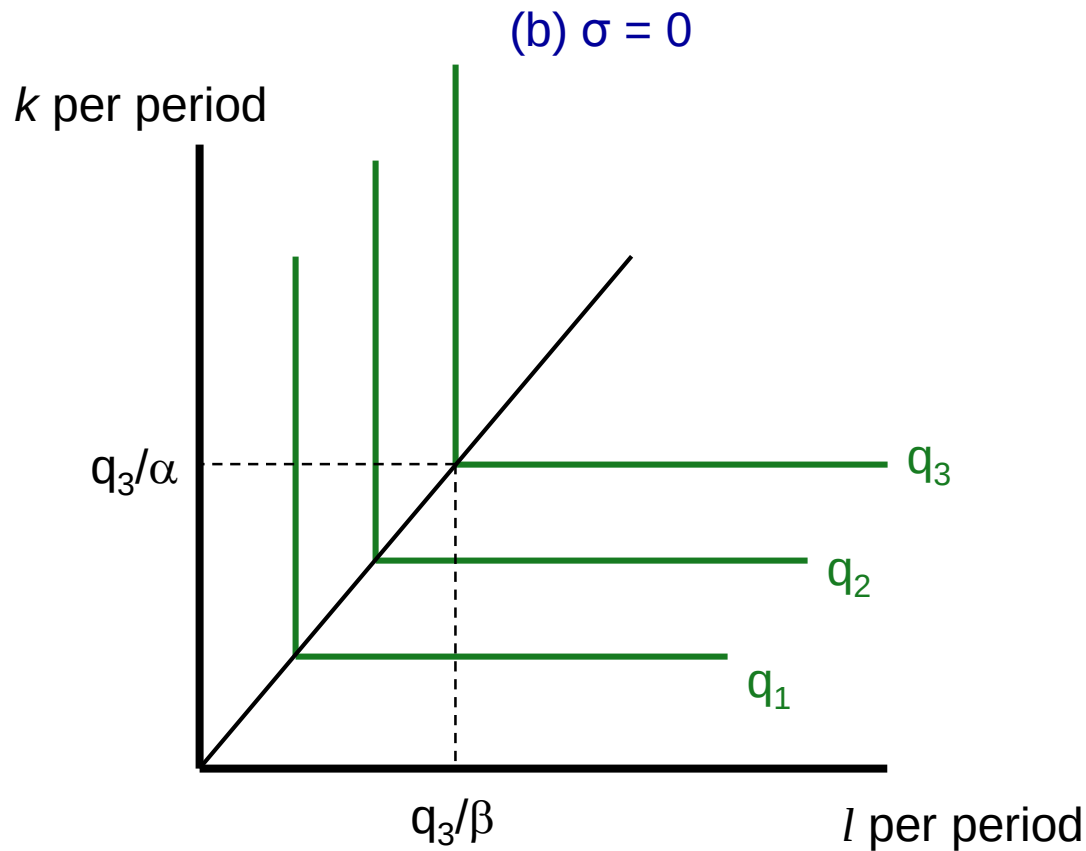
- Fixed proportions production function ($\sigma = 0$):

$$q = \min (\alpha k, \beta l) \quad \alpha, \beta > 0$$

- Capital and labor must always be used in a fixed ratio
 - The firm will always operate along a ray where k/l is constant
- Because k/l is constant, $\sigma = 0$

FIGURE 9.4 (b)

Isoquant Maps for Simple Production Functions with Various Values for σ



Three possible values for the elasticity of substitution are illustrated in these figures.

In (b), the fixed-proportions case, no substitution is possible. The capital-labor ratio is fixed at β/α .

Cobb-Douglas Production Function

- Cobb-Douglas production function ($\sigma = 1$):

$$q = f(k, l) = Ak^{\alpha}l^{\beta} \quad A, \alpha, \beta > 0$$

- This production function can exhibit any returns to scale

$$f(tk, tl) = A(tk)^{\alpha}(tl)^{\beta} = At^{\alpha+\beta}k^{\alpha}l^{\beta} = t^{\alpha+\beta}f(k, l)$$

- if $\alpha + \beta = 1 \Rightarrow$ constant returns to scale
- if $\alpha + \beta > 1 \Rightarrow$ increasing returns to scale
- if $\alpha + \beta < 1 \Rightarrow$ decreasing returns to scale

Cobb-Douglas Production Function

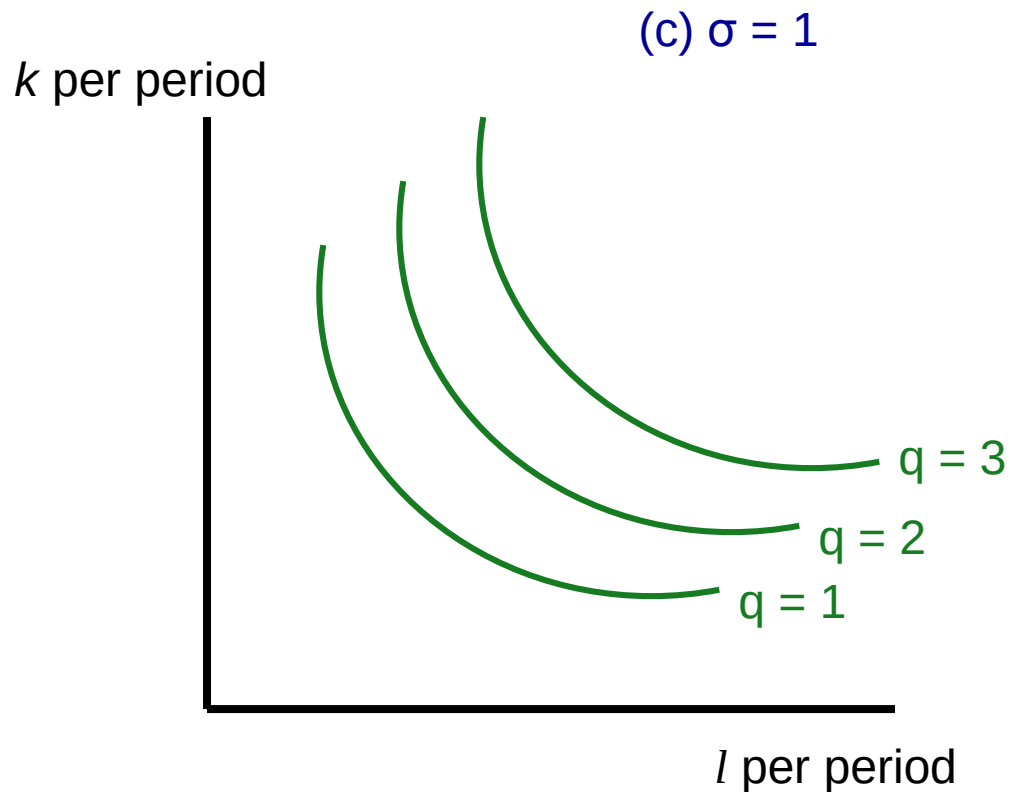
- The Cobb-Douglas production function is linear in logarithms:

$$\ln q = \ln A + \alpha \ln k + \beta \ln l$$

- α is the elasticity of output with respect to k
- β is the elasticity of output with respect to l

FIGURE 9.4 (c)

Isoquant Maps for Simple Production Functions with Various Values for σ



Three possible values for the elasticity of substitution are illustrated in these figures.

A case of limited substitutability is illustrated in (c).

CES Production Function

- CES production function ($\sigma = 1/(1-\rho)$):

$$q = f(k,l) = [k^\rho + l^\rho]^{\gamma/\rho} \quad \rho \leq 1, \rho \neq 0, \gamma > 0$$

- $\gamma > 1 \Rightarrow$ increasing returns to scale
- $\gamma < 1 \Rightarrow$ decreasing returns to scale
- For this production function, $\sigma = 1/(1-\rho)$
 - $\rho = 1 \Rightarrow$ linear production function
 - $\rho = -\infty \Rightarrow$ fixed proportions production function
 - $\rho = 0 \Rightarrow$ Cobb-Douglas production function